

PCAP and EVTX Analysis Report

Week 15 Assignment: Multi-Vector Compromise of seclan.local

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Date of Analysis:	May 8, 2026
Course:	Network Forensics
Assignment:	Week 15 PCAP and EVTX Analysis
Evidence:	FinalExamThreatHunt.zip (SHA1: d14075e4dc87a46c0f8e13ca374adf4a95aa0e09)
Toolset:	Security Onion 2.4.160 (Suricata IDS), Wireshark, NetworkMiner 2.8.1, VirusTotal, Sysmon

1. Abstract

This report investigates a coordinated multi-vector intrusion against the seclan.local Active Directory domain captured on 2021-05-08. Two domain-joined workstations were compromised by a single attacker operating from 192.168.1.10 (a Linux host with the randomly generated hostname Tu1BOSrnxGuOhpcp). The Windows 7 SP1 host at 192.168.1.5 was exploited at 15:29:47 UTC via the EternalBlue (MS17-010) SMBv1 vulnerability, resulting in the delivery of meterpreter.dll over TCP port 4444 and subsequent process injection into the Windows Print Spooler (spoolsv.exe) via CreateRemoteThread. The Windows 10 host at 192.168.1.3 was compromised approximately 90 minutes later via an unauthorized RDP login using Administrator credentials, followed by the delivery of a 26-byte Metasploit payload named pingthing.exe over a raw TCP connection. All findings were corroborated by Suricata IDS alerts in Security Onion, Sysmon Event IDs 8 and 15, NetworkMiner file extraction, and VirusTotal hash analysis. The Meterpreter DLL was confirmed malicious by 56 of 72 antivirus vendors (threat label: trojan.meterpreter/shellcode), and active Metasploit Bind_API constructs were detected on both victim machines.

2. Step A: Network Documentation

The seclan.local environment is a Windows Active Directory domain operating on a Class C LAN segment. Network parameters were derived from analysis of ARP traffic, NetBIOS broadcasts, and the NetworkMiner Hosts tab built from the captured PCAP. The domain controller, DNS server, and gateway were all identified through passive observation of network conversations.

2.1 Network Parameters

LAN Segment Range	192.168.1.0/24
Network	192.168.1.0
Subnet Mask	255.255.255.0
Domain	seclan.local (NetBIOS: SECLAN)
Domain Controller	192.168.1.1 (WIN2008RC2 / WIN2008rc2.seclan.local)
Local DNS Server	192.168.1.1 (WIN2008RC2, port 53)
LAN Gateway	192.168.1.254
LAN Broadcast	192.168.1.255

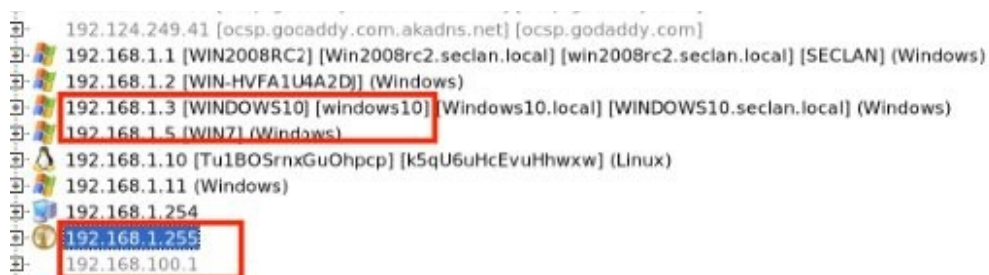


Figure 1: NetworkMiner Hosts tab confirming the network parameters. The gateway at 192.168.1.254 is highlighted, the domain controller WIN2008RC2 sits at 192.168.1.1, the broadcast address is 192.168.1.255, and the attacker host Tu1BOSrnxGuOhpcp (Linux) is at 192.168.1.10.

2.2 Host Inventory

Five hosts were identified as relevant to this investigation. The two domain-joined workstations were both compromised, while the attacker operated from a Linux machine with a randomly generated hostname suggestive of Metasploit Framework default behavior.

IP Address	Hostname	Operating System	Role
192.168.1.1	WIN2008RC2.seclan.local	Windows Server 2008 R2	DC / DNS
192.168.1.3	WINDOWS10.seclan.local	Windows 10	VICTIM
192.168.1.5	WIN7.seclan.local	Windows 7 SP1 (Build 7601)	VICTIM
192.168.1.10	Tu1BOSrnxGuOhpcp	Linux	ATTACKER
192.168.1.254	(Gateway)	N/A	Gateway

3. Step B: Investigation of 192.168.1.3 (WINDOWS10.seclan.local)

3.1 Executive Summary

The Windows 10 workstation at 192.168.1.3 was compromised on 2021-05-08 through an unauthorized Remote Desktop Protocol (RDP) session initiated from the attacker host at 192.168.1.10. NTLM authentication probes against the host began at 15:40 UTC and the successful RDP Administrator login was confirmed by Suricata at 16:15:53. After maintaining the RDP session for approximately 38 minutes, the attacker delivered a 26-byte Metasploit payload named pingthing.exe to the Administrator's Downloads folder at 16:53:15. Sysmon Event ID 15 (FileCreateStreamHash) recorded the file with a Mark of the Web zone identifier confirming browser-initiated download. One second later, Suricata fired ET HUNTING PE EXE Download over raw TCP and ET MALWARE Possible Metasploit Payload Common Construct Bind_API alerts, confirming the file is an active Metasploit bind shell stub. A second wave of identical activity occurred at 16:57 and 16:59, indicating the attacker delivered additional tooling during the same session.

3.2 RDP Initial Access via Administrator Credentials

Filtering Security Onion alerts on source.ip: 192.168.1.3 returned 52 records. The earliest alerts at 15:40:52 show ET INFO NTLMv1 Session Setup Request (Negotiate and Auth phases), indicating the attacker began NTLM authentication probing against the host. This pattern of NTLM challenge-response activity continued in 15-minute intervals consistent with automated credential testing.

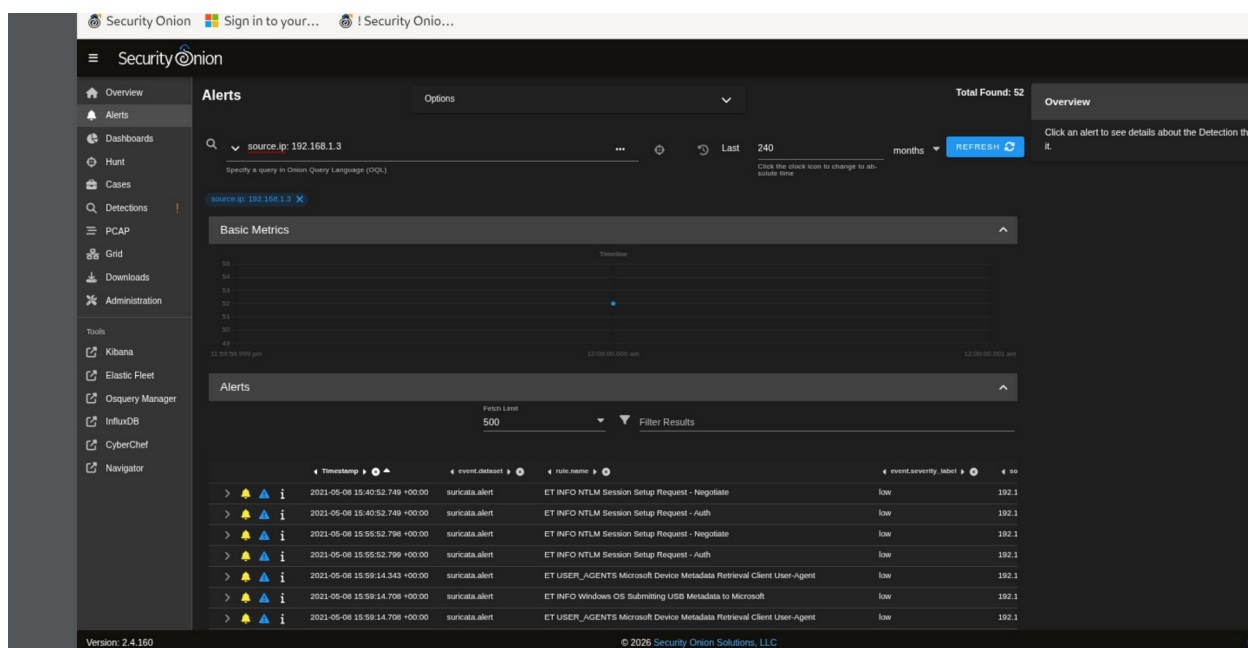


Figure 2: Security Onion Alerts dashboard filtered for source.ip 192.168.1.3 showing the initial NTLM Session Setup probes at 15:40:52, with continued NTLM activity at 15:55:52 and beyond. 52 total alerts triggered for this host.

Reversing the filter to destination.ip: 192.168.1.3 surfaced the critical inbound traffic. At 16:15:53, Suricata fired four consecutive ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request alerts, confirming a successful RDP login using the local Administrator account. RDP login alerts then

continued through the rest of the attack window, indicating a persistent interactive session was maintained on the host.

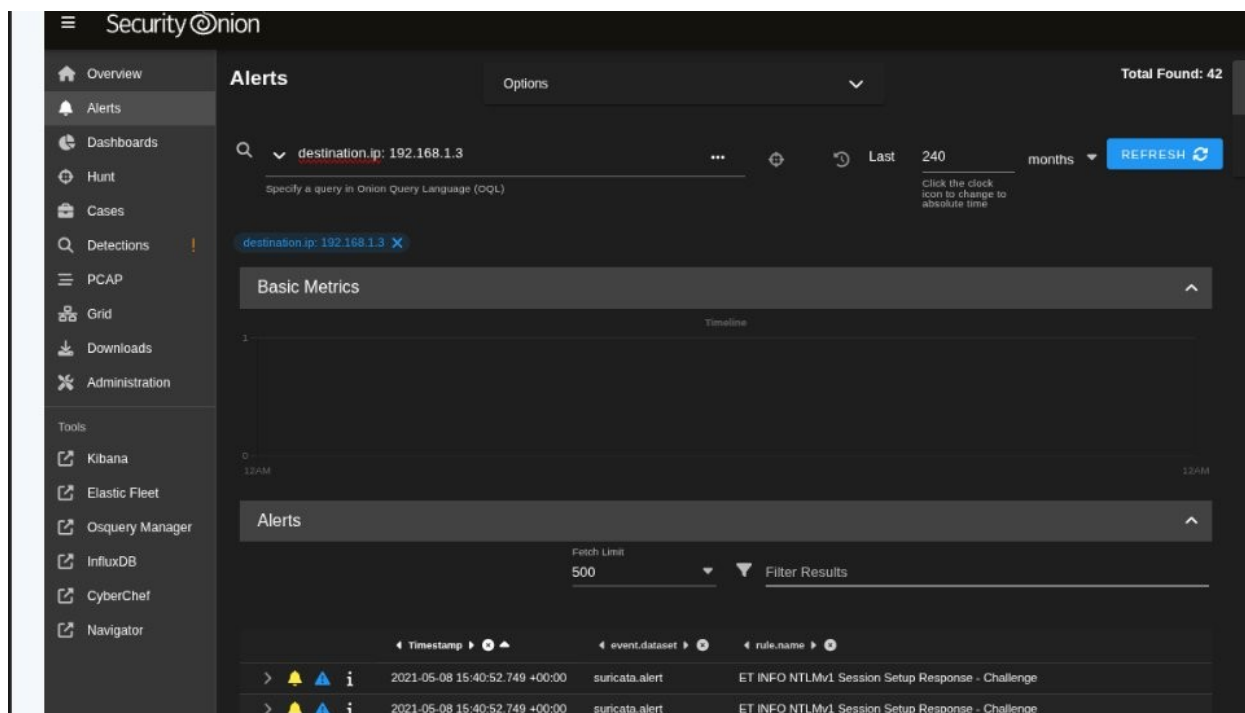


Figure 3: Suricata alerts for traffic destined to 192.168.1.3. The cluster of ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request alerts at 16:15:53 confirms initial access via RDP using the Administrator account.

3.3 Suspicious File Drop: pingthing.exe

Pivoting to Sysmon evidence in Security Onion Hunt, the query `winlog.event_id: 15 (FileCreateStreamHash)` on the `WINDOWS10.seclan.local` host returned 8 events. Sysmon Event ID 15 fires when an alternate data stream (Zone.Identifier, also known as Mark of the Web) is written, which Windows uses to flag files downloaded from the internet. All 8 events were tied to `chrome.exe`, with two distinct PIDs (5664 and 5840) representing two separate download bursts at 16:53 and 16:57.

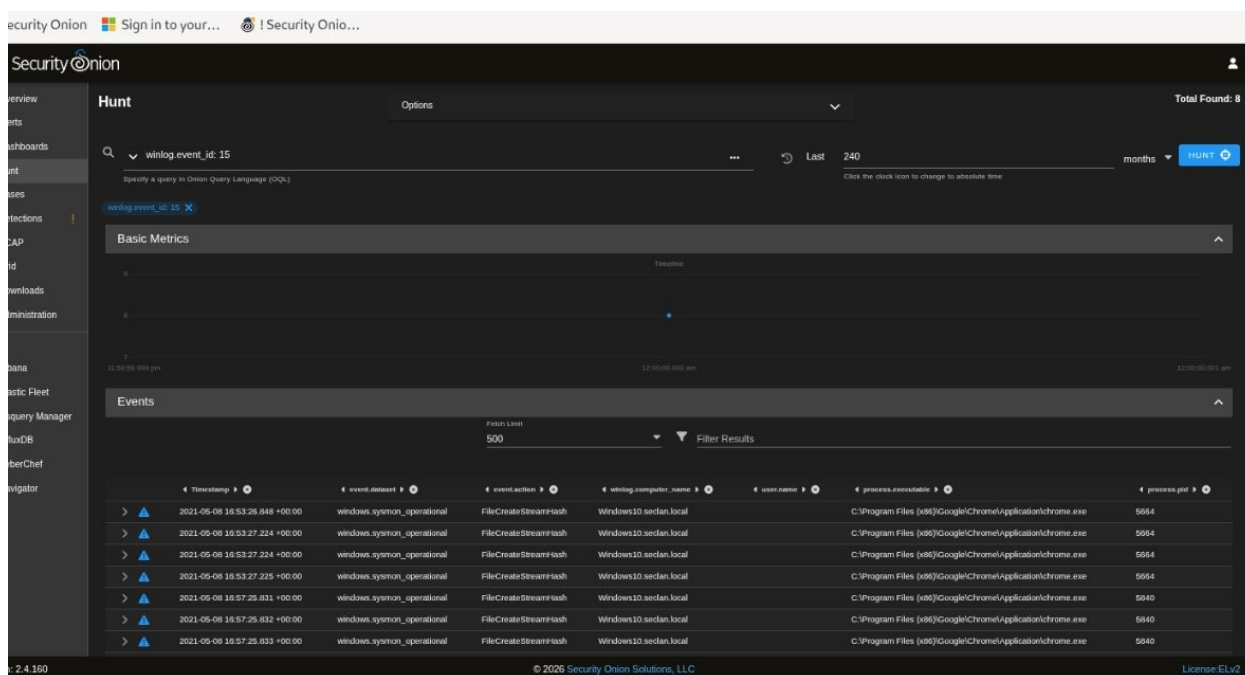


Figure 4: Hunt results for `winlog.event_id: 15` on `Windows10.seclan.local` showing two clusters of `FileCreateStreamHash` events at 16:53:26 (`chrome.exe` PID 5664) and 16:57:25 (`chrome.exe` PID 5840), corresponding to two distinct payload delivery waves.

Expanding the earliest event revealed the suspicious file. Sysmon recorded that `chrome.exe` (PID 5664) created the file `C:\Users\Administrator\Downloads\pingthing.exe` with a `CreationUtcTime` of 2021-05-08 16:53:15.248. The file's hashes were recorded directly in the Sysmon event, providing immediate evidence for hash-based lookup. The deliberately innocuous filename (`pingthing.exe`) is consistent with attacker masquerading techniques designed to avoid suspicion during a quick visual review of Downloads folders.

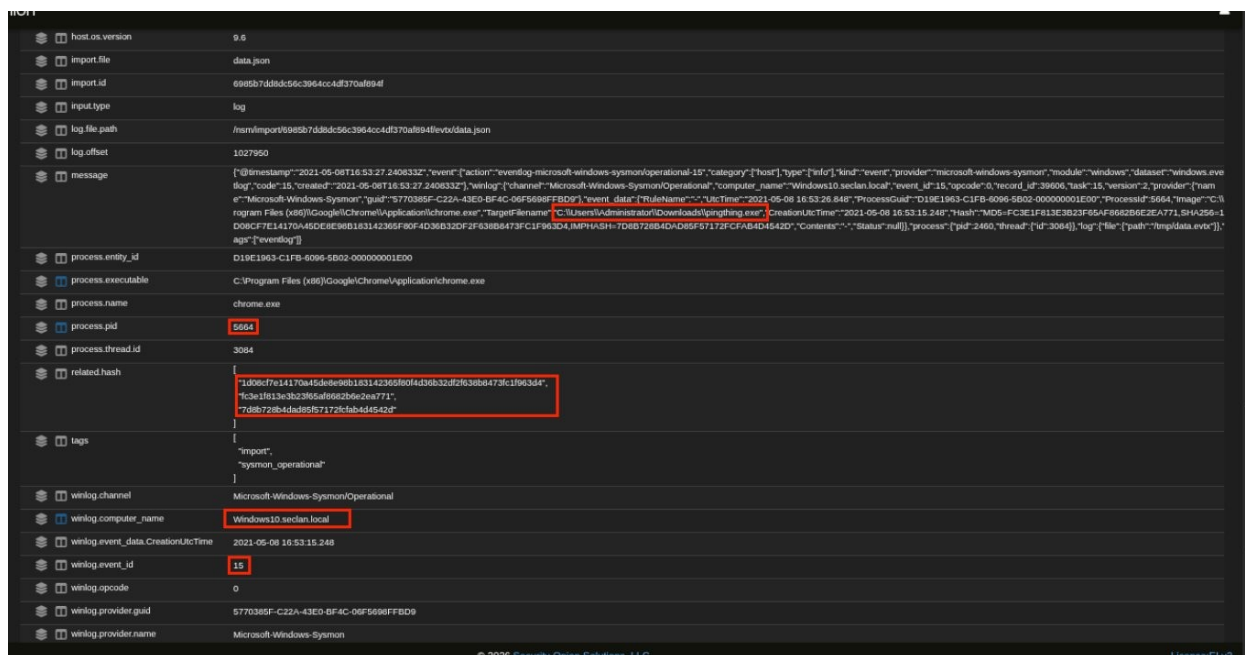


Figure 5: Detailed Sysmon Event ID 15 record. TargetFilename is highlighted as C:\Users\Administrator\Downloads\pingthing.exe, written by chrome.exe (PID 5664). MD5 FC3E1F813E3B23F65AF8682B6E2EA771 and SHA256 1D08CF7E14170A45DE8E98B183142365F80F4D36B32DF2F638B8473FC1F963D4 are recorded directly in the event payload.

3.4 VirusTotal Analysis: pingthing.exe

The pingthing.exe SHA256 hash was queried against VirusTotal. Two facts immediately stood out. First, the file size is only 26 bytes, which is far smaller than any legitimate Windows executable: a normal PE binary's DOS header alone is 64 bytes. A 26-byte executable can only be a stub or shellcode container. Second, despite zero antivirus vendors flagging the file, the community score is -76, indicating multiple analysts manually marked it as malicious. The combination of zero AV detections with a deeply negative community score is a well-documented signature of a custom Metasploit payload designed to evade signature-based detection.

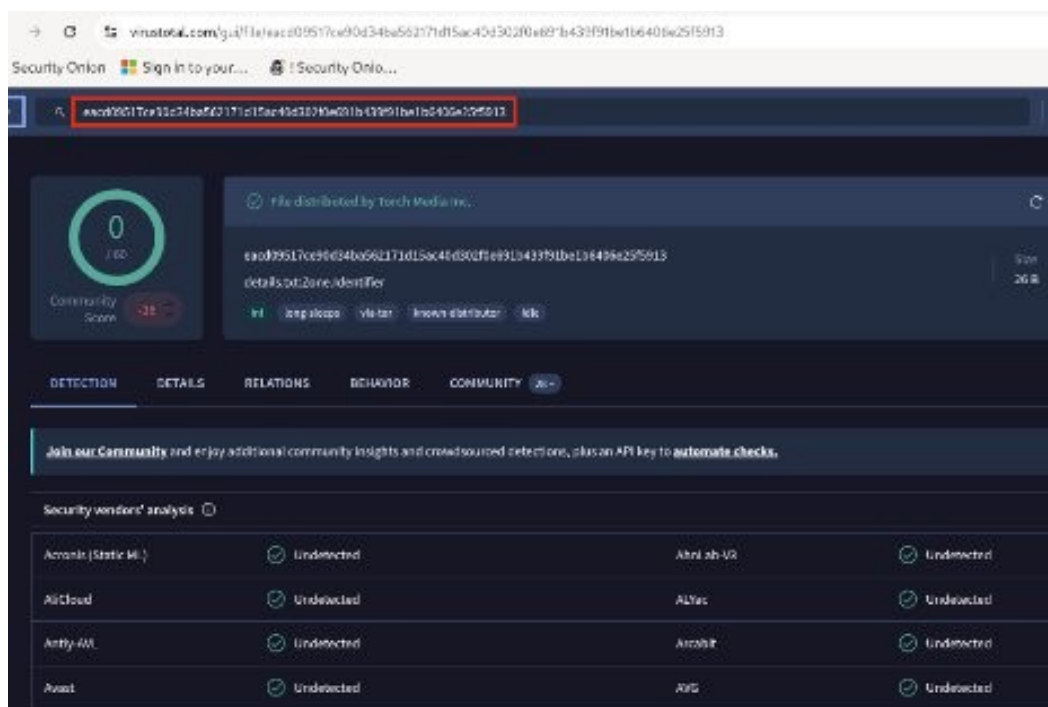
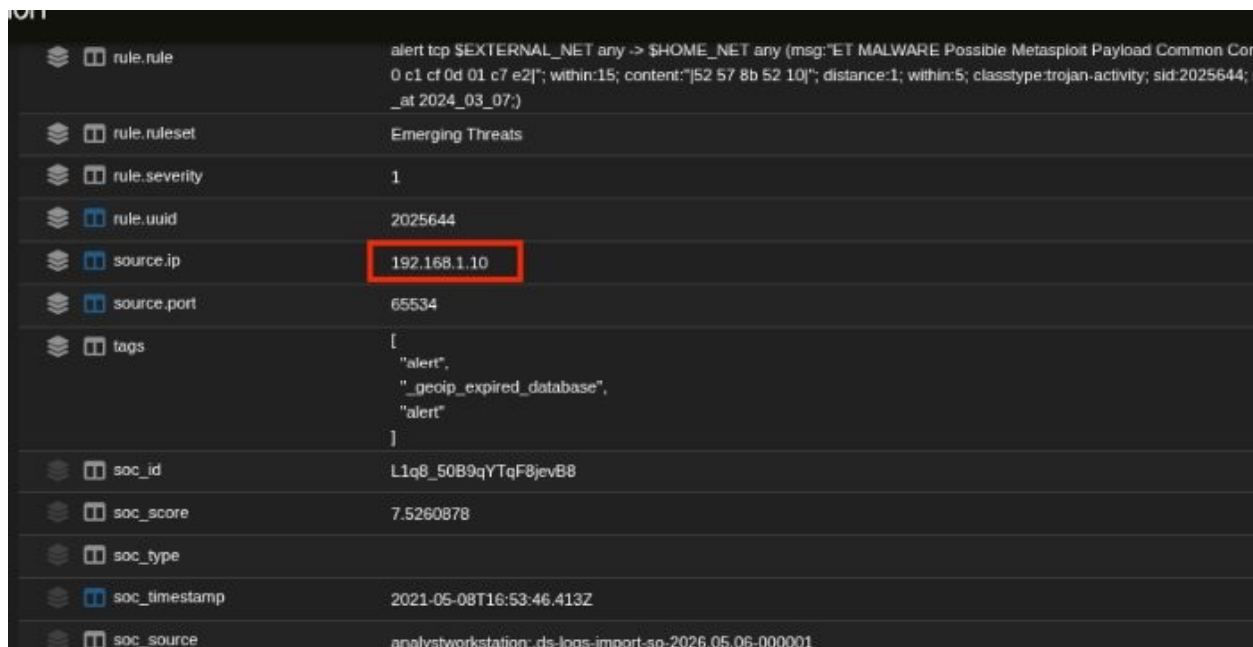


Figure 6: VirusTotal analysis of pingthing.exe. 0 of 92 vendors flagged the file as malicious, but the file size of 26 bytes, the deeply negative community score, and tags including via-tar and known-distributor are consistent with a custom Metasploit payload stub designed to evade signature-based detection.

3.5 Suricata Confirmation of Metasploit Payload

While Sysmon recorded the file landing on disk, the Suricata IDS confirmed the underlying delivery method and payload identity. At 16:53:45, Suricata fired the ET HUNTING PE EXE Download over raw TCP rule, indicating the file was delivered over a raw TCP socket rather than HTTP or HTTPS. This explains why HTTP-layer Wireshark filters (such as http.request.uri contains ".exe") returned only legitimate Windows Update traffic. Raw TCP delivery is a deliberate evasion technique that bypasses application-layer inspection.

One second later, at 16:53:46, the high-severity rule ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) fired with source IP 192.168.1.10 and destination 192.168.1.3. This signature matches the byte pattern used by Metasploit's bind shell payloads during the initial Windows API resolution phase, confirming the payload is an active Metasploit Bind shell. Suricata rule SID 2025644 from the Emerging Threats ruleset triggered this match.



rule.rule	alert tcp \$EXTERNAL_NET any -> \$HOME_NET any (msg:"ET MALWARE Possible Metasploit Payload Common Construct Bind_API"; within:15; content:"[52 57 8b 52 10]"; distance:1; within:5; classtype:trojan-activity; sid:2025644; rev:1; _at 2024_03_07;)
rule.ruleset	Emerging Threats
rule.severity	1
rule.uuid	2025644
source.ip	192.168.1.10
source.port	65534
tags	["alert", "_geoip_expired_database", "alert"]
soc_id	L1q8_50B9qYTqF8jevB8
soc_score	7.5260878
soc_type	
soc_timestamp	2021-05-08T16:53:46.413Z
soc_source	analystworkstation:ds-logs-import-so-2026.05.06-000001

Figure 7: Detailed view of the ET MALWARE Possible Metasploit Payload Common Construct Bind_API alert. Source IP 192.168.1.10 (the attacker host), rule SID 2025644, severity 1 (high). Triggered at 2021-05-08 16:53:46 UTC, one second after the PE EXE download alert.

Scrolling through the full alerts list for 192.168.1.3 reveals the complete attack chain in chronological order. The four-alert RDP login cluster at 16:15:53 marks initial access, the NTLM session activity at 16:25 and 16:52 represents the active RDP session, and the two payload delivery waves at 16:53 and 16:59 each produce paired ET HUNTING PE EXE Download over raw TCP and ET MALWARE Metasploit Bind_API alerts.

>		2021-05-08 16:15:53.556 +00:00	suricata.alert	ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request
>		2021-05-08 16:15:53.556 +00:00	suricata.alert	ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request
>		2021-05-08 16:15:53.560 +00:00	suricata.alert	ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request
>		2021-05-08 16:15:53.560 +00:00	suricata.alert	ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request
>		2021-05-08 16:25:52.896 +00:00	suricata.alert	ET INFO NTLMv1 Session Setup Response - Challenge
>		2021-05-08 16:25:52.896 +00:00	suricata.alert	ET INFO NTLMv1 Session Setup Response - Challenge
>		2021-05-08 16:25:52.896 +00:00	suricata.alert	ET INFO NTLMv1 Session Setup Response - Challenge
>		2021-05-08 16:52:09.647 +00:00	suricata.alert	ET INFO NTLMv1 Session Setup Response - Challenge
>		2021-05-08 16:52:09.647 +00:00	suricata.alert	ET INFO NTLMv1 Session Setup Response - Challenge
>		2021-05-08 16:53:45.708 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP
>		2021-05-08 16:53:45.708 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP
>		2021-05-08 16:53:45.708 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP
>		2021-05-08 16:53:46.413 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)
>		2021-05-08 16:53:46.413 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)
>		2021-05-08 16:53:46.413 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)
>		2021-05-08 16:59:08.806 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP
>		2021-05-08 16:59:08.806 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP
>		2021-05-08 16:59:08.806 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP
>		2021-05-08 16:59:09.201 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)
>		2021-05-08 16:59:09.201 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)

Figure 8: Complete chronological alert sequence for the 192.168.1.3 attack chain. Four RDP Administrator Login Request alerts at 16:15:53 (initial access), NTLMv1 session activity at 16:25 and 16:52 (active session), and two waves of ET HUNTING PE EXE Download over raw TCP + ET MALWARE Metasploit Bind_API alerts at 16:53:45-46 and 16:59:08-09 (payload delivery).

3.6 Second Payload Delivery Wave at 16:57 to 16:59

A second cluster of Sysmon Event ID 15 events fired at 16:57:25 with chrome.exe PID 5840, indicating the attacker delivered additional files during the same RDP session. Suricata confirmed this with another set of ET HUNTING PE EXE Download over raw TCP and ET MALWARE Metasploit Bind_API alerts at 16:59. This pattern suggests the attacker was iterating on tooling, possibly delivering an upgraded payload after testing the initial drop.

3.7 Timeline Summary: 192.168.1.3

Timestamp (UTC)	Event	Severity
15:40:52	ET INFO NTLMv1 Session Setup Request (Negotiate and Auth) probes begin from 192.168.1.10	LOW
15:55:52	Continued NTLM challenge-response activity (credential testing pattern)	LOW
16:10:52	Recurring NTLM Session Setup probes consistent with automated authentication testing	LOW
16:15:53	ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request fires (RDP login confirmed)	HIGH

16:25:52	Post-login NTLM Session Setup responses observed (session establishment)	LOW
16:52:09	Renewed NTLM challenge activity during active RDP session	LOW
16:53:15	Sysmon Event ID 15: pingthing.exe (26 bytes) created in C:\Users\Administrator\Downloads via chrome.exe (PID 5664)	HIGH
16:53:45	ET HUNTING PE EXE Download over raw TCP (delivery method confirmed)	MED
16:53:46	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) from 192.168.1.10	HIGH
16:57:25	Second Sysmon Event ID 15 cluster: additional download via chrome.exe (PID 5840)	MED
16:59:08	Second wave of ET HUNTING PE EXE Download over raw TCP alerts	HIGH
16:59:09	Second ET MALWARE Metasploit Bind_API alert from 192.168.1.10	HIGH

3.8 Chronological Listing of Evidence: 192.168.1.3

Evidence 1: RDP Administrator Login Request (Suricata Alert)

Timestamp	2021-05-08 16:15:53 UTC
Source	Suricata IDS via Security Onion
Rule	ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request
Source IP	192.168.1.10 (Tu1BOSrnxGuOhpcp, Linux)
Destination IP	192.168.1.3 (WINDOWS10.seclan.local)
Significance	Confirmed initial access via RDP using the local Administrator account. This is the entry point into the Windows 10 host.
MITRE ATT&CK	T1078 (Valid Accounts) T1021.001 (RDP)

Evidence 2: Sysmon FileCreateStreamHash for pingthing.exe

Timestamp	2021-05-08 16:53:15.248 UTC
Event ID	15 (Sysmon FileCreateStreamHash)
Host	WINDOWS10.seclan.local
Process	C:\Program Files (x86)\Google\Chrome\Application\chrome.exe (PID 5664)
Target File	C:\Users\Administrator\Downloads\pingthing.exe
File Size	26 bytes
MD5	FC3E1F813E3B23F65AF8682B6E2EA771
SHA256	1D08CF7E14170A45DE8E98B183142365F80F4D36B32DF2F638B8473FC1F963D4

Zone Identifier	Present (Mark of the Web)
Significance	Suspicious file landed in Administrator's Downloads folder via Chrome. Filename appears innocuous (pingthing.exe), but the 26-byte size is impossible for a legitimate PE.
MITRE ATT&CK	T1105 (Ingress Tool Transfer) T1036 (Masquerading)

Evidence 3: VirusTotal Hash Analysis for pingthing.exe

Hash Queried	1D08CF7E14170A45DE8E98B183142365F80F4D36B32DF2F638B8473FC1F963D4
Detection Ratio	0 / 92 vendors
File Size	26 bytes
Community Score	-76 (community-flagged as suspicious)
Significance	Combination of 0 AV detections, deeply negative community score, and 26-byte size is consistent with a custom Metasploit payload designed to bypass signature-based detection. The file is too small to be a legitimate executable.

Evidence 4: ET HUNTING PE EXE Download over Raw TCP

Timestamp	2021-05-08 16:53:45 UTC
Rule	ET HUNTING PE EXE Download over raw TCP
Source IP	192.168.1.10
Destination IP	192.168.1.3
Significance	Confirms pingthing.exe was delivered over a raw TCP socket rather than HTTP/HTTPS. This explains why HTTP-layer Wireshark filters did not capture the transfer. Raw TCP delivery is a deliberate evasion of application-layer inspection.
MITRE ATT&CK	T1571 (Non-Standard Port) T1105 (Ingress Tool Transfer)

Evidence 5: ET MALWARE Possible Metasploit Payload Bind_API

Timestamp	2021-05-08 16:53:46 UTC
Rule	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)
Rule SID	2025644
Source IP	192.168.1.10
Destination IP	192.168.1.3
Severity	High
Significance	Definitive confirmation that pingthing.exe is a Metasploit bind shell payload. The Bind_API signature matches the byte pattern Metasploit shellcode uses

	during initial Windows API resolution. Combined with the 26-byte file and raw TCP delivery, this confirms an active Metasploit foothold.
MITRE ATT&CK	T1059.003 (Windows Command Shell) T1055 (Process Injection)

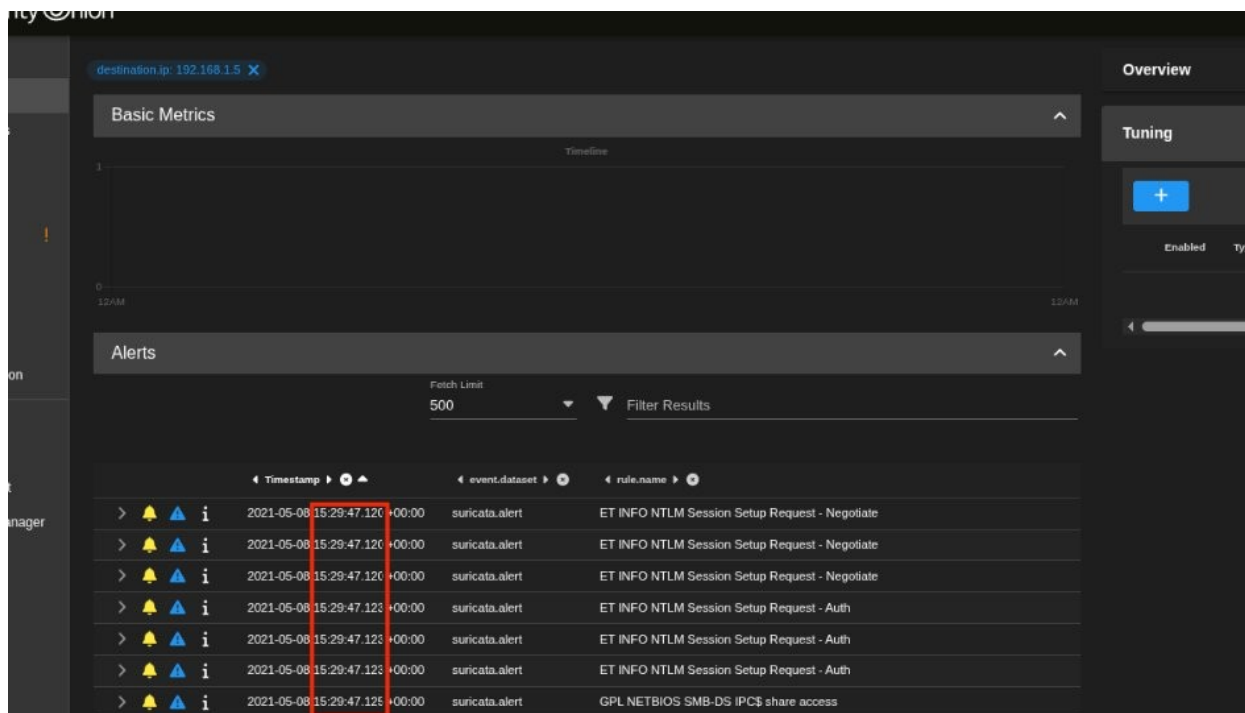
4. Step C: Investigation of 192.168.1.5 (WIN7.seclan.local)

4.1 Executive Summary

The Windows 7 SP1 workstation at 192.168.1.5 was compromised on 2021-05-08 via the EternalBlue exploit (MS17-010), a critical pre-authentication remote code execution vulnerability in the Windows SMBv1 driver. At 15:29:47 UTC, the attacker at 192.168.1.10 began probing the SMB service on WIN7. Suricata fired multiple high-severity EternalBlue probe alerts (both MSF-style and Generic Flags signatures), confirming the attacker used the Metasploit Framework. Eleven seconds later at 15:29:58, the ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010 rule fired, confirming successful exploitation. Within the same second, NetworkMiner captured the delivery of meterpreter.dll (200,262 bytes) from 192.168.1.10 to 192.168.1.5 over TCP port 4444, the default Metasploit listener port. VirusTotal confirmed the DLL as malicious by 56 of 72 vendors with the threat label trojan.meterpreter/shellcode. The Meterpreter agent then injected itself into the Windows Print Spooler service (spoolsv.exe) via CreateRemoteThread at 15:30:13, achieving process migration into a persistent, trusted SYSTEM-privileged process. Post-exploitation activity included SAMR and SRVSVC enumeration of the domain controller at 192.168.1.1 and outbound connections to file.io (18.232.47.250).

4.2 EternalBlue Exploit (MS17-010)

Filtering Security Onion alerts on destination.ip: 192.168.1.5 returned alerts beginning with GPL NETBIOS SMB-DS IPC\$ Share Access at 15:29:47, indicating the attacker connected to the IPC\$ administrative share. Within the same second, two EternalBlue probe signatures fired in parallel: ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (MSF style) and the same probe with Generic Flags. The MSF-style signature matches Metasploit's specific transaction fingerprint, while the Generic Flags rule matches the broader EternalBlue protocol anomaly. Together these confirm both the exploit identity and the framework used to deliver it.



	Timestamp	event.dataset	rule.name
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.120 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Negotiate
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.120 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Negotiate
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.120 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Negotiate
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.123 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Auth
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.123 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Auth
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.123 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Auth
> 🚨 🔵 ⓘ	2021-05-08 15:29:47.125 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access

Figure 9: Initial Suricata alerts filtered on destination.ip 192.168.1.5. The pattern begins at 15:29:47 with NTLM Session Setup Request - Negotiate and Auth alerts, followed by GPL NETBIOS SMB-DS IPC\$ share access alerts as the attacker establishes the SMB connection used for EternalBlue delivery.

Eleven seconds later at 15:29:58, the ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010 high-severity rule fired, indicating the M2 exploitation stage executed successfully. This is the moment the attacker achieved kernel-mode remote code execution on WIN7 without any user interaction required. The exploit leverages CVE-2017-0144, the vulnerability publicly disclosed in 2017 after the NSA's EternalBlue tool was leaked by the Shadow Brokers group.

>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.123 +00:00	suricata.alert	ET INFO NTLM Session Setup Request - Auth	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.125 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.125 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.125 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.126 +00:00	suricata.alert	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (MSF style)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.126 +00:00	suricata.alert	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (Generic Flags)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.126 +00:00	suricata.alert	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (MSF style)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.126 +00:00	suricata.alert	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (Generic Flags)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.126 +00:00	suricata.alert	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (MSF style)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:47.126 +00:00	suricata.alert	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (Generic Flags)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.221 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.221 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.221 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.286 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.286 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.286 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:48.286 +00:00	suricata.alert	GPL NETBIOS SMB-DS IPC\$ share access	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.385 +00:00	suricata.alert	ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.385 +00:00	suricata.alert	ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.385 +00:00	suricata.alert	ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.539 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.539 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.539 +00:00	suricata.alert	ET HUNTING PE EXE Download over raw TCP	low
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.540 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.540 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	high
>	🔔	🔵	🔵	🔵	2021-05-08 15:29:58.540 +00:00	suricata.alert	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	high

Figure 10: Critical Suricata alert sequence for 192.168.1.5. The high-severity ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 alerts fire at 15:29:47, followed eleven seconds later by ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010 at 15:29:58. Within the same second, ET HUNTING PE EXE Download over raw TCP and ET MALWARE Possible Metasploit Payload Common Construct Bind_API alerts confirm payload delivery.

4.3 Meterpreter DLL Delivery via TCP 4444

Switching to NetworkMiner and filtering the Files tab on 192.168.1.5 surfaced the most damning piece of evidence in the entire investigation. At frame 590 of the PCAP, NetworkMiner captured a file named meterpreter.dll (200,262 bytes / 195.57 KB) being transferred from 192.168.1.10 to 192.168.1.5 over TCP port 4444. TCP 4444 is the default listener port used by the Metasploit Framework, and the file was named meterpreter.dll without any attempt at obfuscation, indicating the attacker did not bother concealing the tooling.

Frame nr.	Filename	Extension	Size	Source host	S. port	Destination host
590	meterpreter.dll	dll	200 262 B	192.168.1.10 [Tu1BOSrnxGuOhpcp] [k5qU6uHc...	TCP 4444	192.168.1.5 [WIN7] (Windows)
13155	samr	samr	524 B	192.168.1.5 [WIN7] (Windows)	TCP 58833	192.168.1.1 [WIN2008R2] (Windows)
13487	gpt.ini	ini	23 B	192.168.1.1 [WIN2008R2] (Windows)	TCP 445	192.168.1.5 [WIN7] (Windows)
78420	srsvcs	srsvcs	160 B	192.168.1.5 [WIN7] (Windows)	TCP 57901	192.168.1.1 [WIN2008R2] [WIN2008R2.seclan....
78689	gpt[1].ini	ini	23 B	192.168.1.1 [WIN2008R2] [WIN2008R2.seclan....	TCP 445	192.168.1.5 [WIN7] (Windows)
79031	gpt[2].ini	ini	23 B	192.168.1.1 [WIN2008R2] [WIN2008R2.seclan....	TCP 445	192.168.1.5 [WIN7] (Windows)
79828	index.html	html	82 B	18.232.47.250 [file.io]	TCP 80	192.168.1.5 [WIN7] (Windows)
81086	file.io[2].cer	cer	1 383 B	18.232.47.250 [file.io]	TCP 443	192.168.1.5 [WIN7] (Windows)
81086	Amazon[2].cer	cer	1 101 B	18.232.47.250 [file.io]	TCP 443	192.168.1.5 [WIN7] (Windows)
81086	Amazon Root CA 1[2].cer	cer	1 174 B	18.232.47.250 [file.io]	TCP 443	192.168.1.5 [WIN7] (Windows)
81086	Starfield Services Root Cert[2].cer	cer	1 145 B	18.232.47.250 [file.io]	TCP 443	192.168.1.5 [WIN7] (Windows)

Figure 11: NetworkMiner Files tab filtered for 192.168.1.5. Frame 590 shows meterpreter.dll (200,262 bytes) delivered from 192.168.1.10 [Tu1BOSrnxGuOhpcp] over TCP port 4444. Subsequent rows show post-exploitation traffic: samr and srsvcs protocols against the domain controller at 192.168.1.1, plus outbound connections to file.io (18.232.47.250).

Clicking on the meterpreter.dll entry in NetworkMiner exposed the full file metadata. The Last Write Time of 5/8/2021 3:29 PM matches the EternalBlue exploit timestamp, the source is confirmed as 192.168.1.10, and the destination is WIN7. NetworkMiner automatically extracted the file to disk for offline analysis and computed its MD5 hash.

meterpreter.dll - File Details	
Destination	192.168.1.5 [WIN7] (Windows)
LastWriteTime	5/8/2021 3:29 PM
MD5	bb8192f17603c8098f6d9985f1bd1713
Name	meterpreter.dll
Path	/opt/networkminer/AssembledFiles/192.168.1.10/
SHA1	27e851e0c786a5d9605fc10bfec8d12ddcfcd9d
SHA256	f471a1873cc2d793e2db1efcc6b0789401c72facb6d
Size	200262
Source	192.168.1.10 [Tu1BOSrnxGuOhpcp] [k5qU6uHcEv]

Figure 12: NetworkMiner File Details panel for meterpreter.dll. Destination 192.168.1.5 (WIN7), MD5 bb8192f17603c8098f6d9985f1bd1713, source 192.168.1.10, file size 200,262 bytes. Last Write Time 5/8/2021 3:29 PM aligns with the EternalBlue exploit timeline.

4.4 VirusTotal Analysis: meterpreter.dll

Submitting the meterpreter.dll SHA256 hash to VirusTotal returned a definitive malicious classification. 56 of 72 vendors flagged the file with detections including trojan.meterpreter, Backdoor.Win.Meterpreter, Exp.Win.Meterpreter, HackTool:Win32/Meterpreter, Win32:Metasploit-C, and Generic.ShellCode.Marte. The popular threat label assigned by VirusTotal is trojan.meterpreter/shellcode and the family labels are meterpreter, shellcode, and marte. Behavioral

tags include checks-user-input, long-sleeps, detect-debug-environment, and 64bits, all consistent with anti-analysis behavior typical of Meterpreter staged payloads.

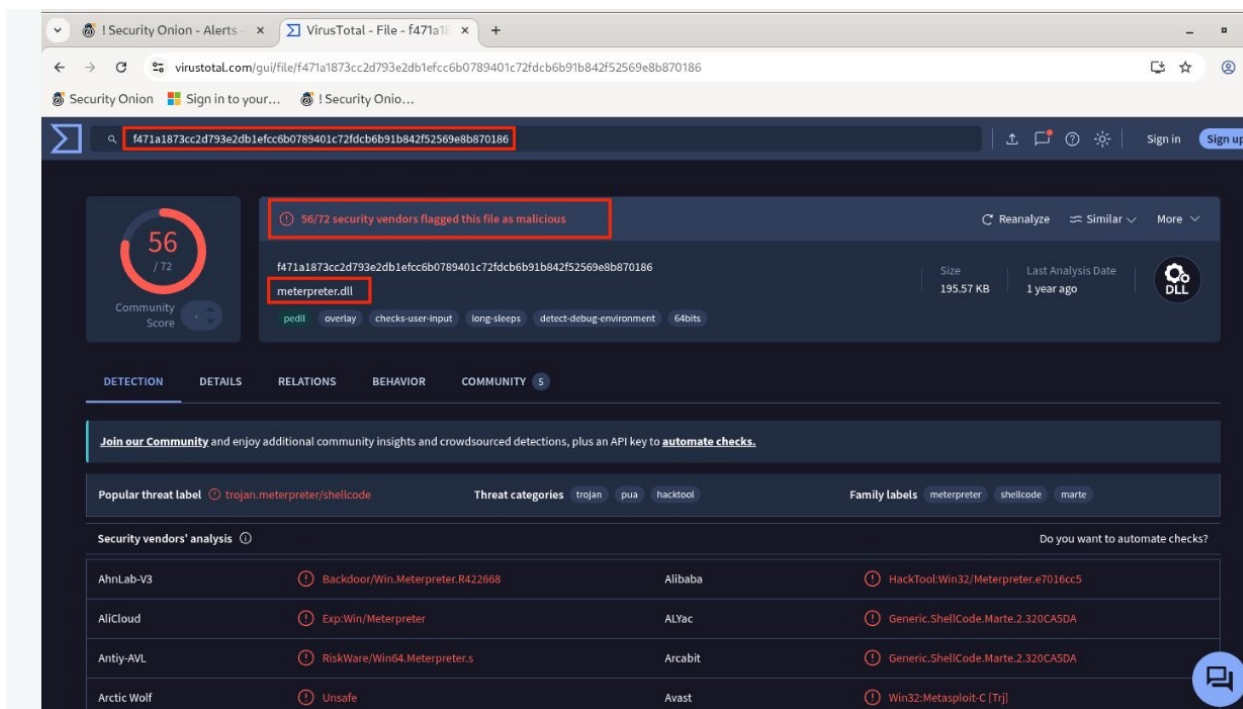


Figure 13: VirusTotal analysis of meterpreter.dll (SHA256:

f471a1873cc2d793e2db1efcc6b0789401c72fdb6b91b842f52569e8b870186). 56 of 72 vendors flagged the file as malicious. Threat label trojan.meterpreter/shellcode with family labels meterpreter, shellcode, marte. Multiple vendors specifically identified the file as Meterpreter or Metasploit-C.

4.5 Process Injection into spoolsv.exe via CreateRemoteThread

Following meterpreter.dll delivery, the attacker invoked Meterpreter's process migration command to inject the Meterpreter agent into a more persistent host process. Sysmon Event ID 8 (CreateRemoteThread) records every instance of one process creating a thread inside another process, which is the primary technique used for process injection. The Hunt query winlog.event_id: 8 returned 17 events on WIN7. Filtering out the legitimate Sysmon64.exe baseline activity isolated two suspicious clusters: spoolsv.exe (PID 1092) creating remote threads at 15:30:13 and again at 15:31:05.

spoolsv.exe is the Windows Print Spooler service. It runs continuously as SYSTEM, making it an attractive migration target because the injected Meterpreter agent inherits SYSTEM privileges and survives even if the original exploit shellcode process is terminated. Print Spooler injection is a well-documented post-exploitation technique used by Meterpreter, Cobalt Strike, and other commodity C2 frameworks.

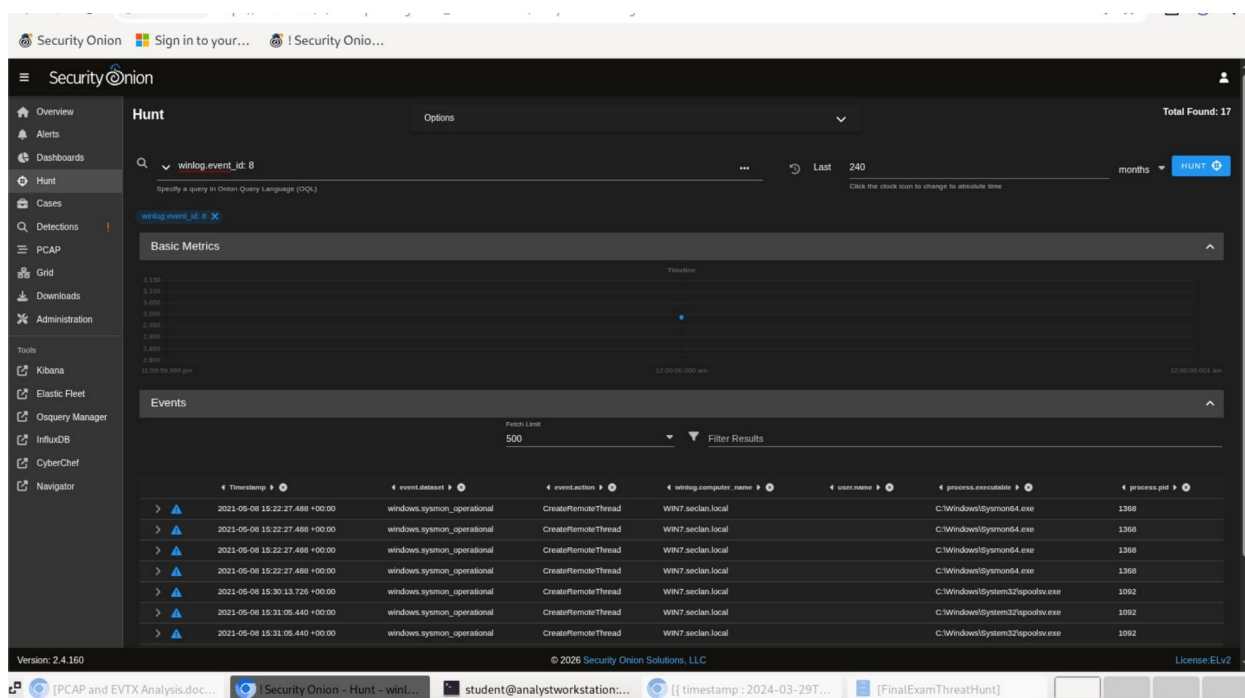


Figure 14: Hunt query for `winlog.event_id: 8` (Sysmon `CreateRemoteThread`) on `WIN7.seclan.local`. `Sysmon64.exe` events at 15:22:27 represent legitimate baseline activity (Sysmon's own monitoring threads). The `spoolsv.exe` (PID 1092) events at 15:30:13 and 15:31:05 represent Meterpreter process injection into the Windows Print Spooler service.

4.6 Host Confirmation via Sysmon Registry Events

To confirm the WIN7 hostname mapping, the Hunt query `winlog.event_id: 12 OR 13 OR 14` was run, returning Sysmon registry events from the host. The expanded event payload showed `winlog.computer_name: WIN7.seclan.local` and `winlog.event_data.BuildVersion: 7601` (the build number for Windows 7 Service Pack 1), confirming the hostname to IP mapping. The same workflow was used to confirm Windows10.seclan.local at 192.168.1.3.

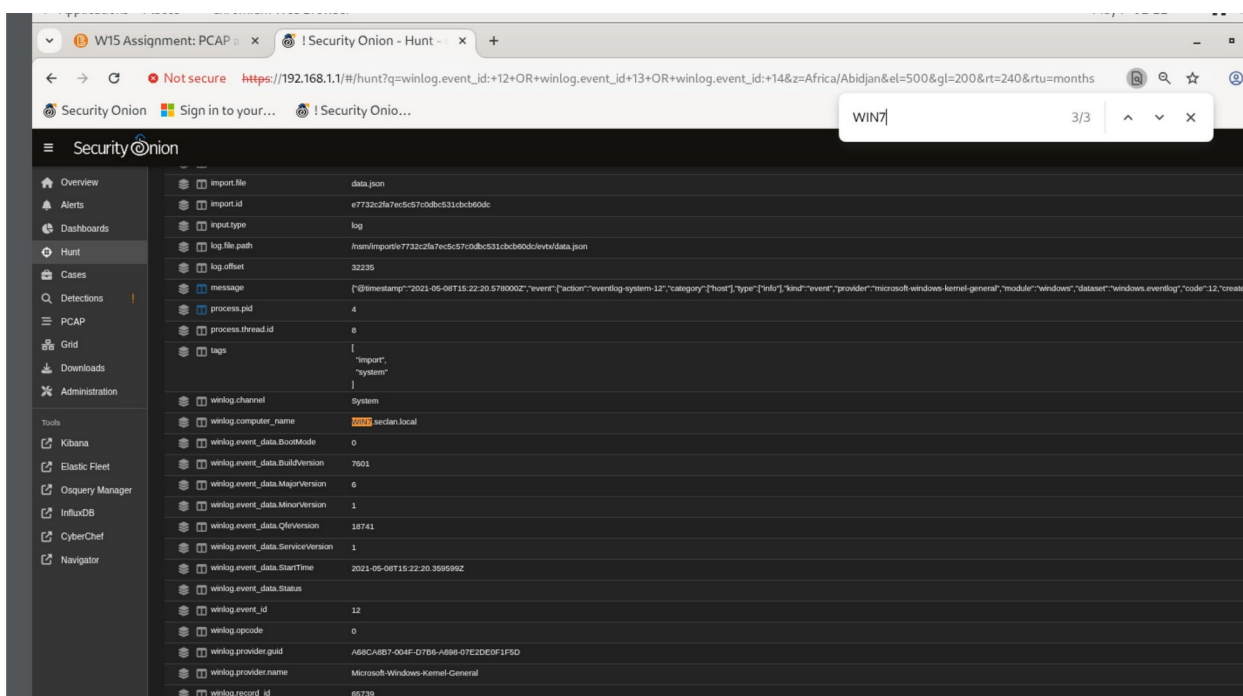


Figure 15: Sysmon registry event detail confirming winlog.computer_name as WIN7.seclan.local with BuildVersion 7601 (Windows 7 SP1). This mapping was established as part of host attribution during the investigation.

4.7 Post-Exploitation Reconnaissance

Following successful Meterpreter injection, the attacker invoked Meterpreter's built-in reconnaissance modules against the domain controller. NetworkMiner captured two SMB-based RPC interactions from WIN7 to 192.168.1.1 (WIN2008RC2):

- **samr (Security Account Manager Remote):** Used to enumerate domain user accounts, group memberships, and Security Identifiers. SAMR queries are typical of the post/multi/recon modules invoked early in a Meterpreter session.
- **srvsvc (Server Service Remote):** Used to enumerate network shares and active sessions on the target. SRVSVC enumeration helps the attacker identify which hosts the DC connects to and which administrative shares are exposed.

Additionally, WIN7 made multiple TCP 443 connections to 18.232.47.250, which resolves to file.io, a public anonymous file-sharing platform. Connections to file-sharing services from a compromised host are a strong indicator of either data staging (downloading additional tooling) or data exfiltration.

4.8 Timeline Summary: 192.168.1.5

Timestamp (UTC)	Event	Severity
15:29:47	GPL NETBIOS SMB-DS IPC\$ Share Access (attacker connects to IPC\$)	LOW
15:29:47	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (MSF style) from 192.168.1.10	HIGH
15:29:47	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (Generic Flags)	HIGH
15:29:48	Continued IPC\$ share access (exploit	LOW

	channel established)	
15:29:58	ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010 (kernel RCE achieved)	HIGH
15:29:58	ET HUNTING PE EXE Download over raw TCP (payload delivery)	HIGH
15:29:58	Frame 590: meterpreter.dll (200,262 bytes) delivered from 192.168.1.10 via TCP 4444	HIGH
15:29:58	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	HIGH
15:30:13	Sysmon Event ID 8: spoolsv.exe (PID 1092) CreateRemoteThread (process injection)	HIGH
15:31:05	Additional Sysmon Event ID 8 events from spoolsv.exe (continued injection)	HIGH
Post-exploit	samr protocol from WIN7 to DC at 192.168.1.1 (domain user enumeration)	MED
Post-exploit	srvsvc protocol from WIN7 to DC at 192.168.1.1 (share enumeration)	MED
Post-exploit	TCP 443 connections from WIN7 to 18.232.47.250 (file.io)	MED

4.9 Chronological Listing of Evidence: 192.168.1.5

Evidence 1: EternalBlue Probe Alerts (Suricata)

Timestamp	2021-05-08 15:29:47 UTC
Rules Fired	ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (MSF style) + ET EXPLOIT Possible ETERNALBLUE Probe MS17-010 (Generic Flags)
Severity	High (both rules)
Source IP	192.168.1.10
Destination IP	192.168.1.5
Significance	Two distinct EternalBlue probe signatures fired simultaneously, confirming the attacker used the Metasploit Framework's EternalBlue module against an unpatched Windows 7 SP1 host.
MITRE ATT&CK	T1190 (Exploit Public-Facing Application) CVE-2017-0144 (MS17-010)

Evidence 2: EternalBlue Exploit Execution

Timestamp	2021-05-08 15:29:58 UTC
Rule	ET EXPLOIT ETERNALBLUE Exploit M2 MS17-010
Severity	High
Source IP	192.168.1.10
Destination IP	192.168.1.5

Significance	The M2 stage of the EternalBlue exploit (the kernel-mode payload). This is the moment the attacker achieved remote code execution on WIN7 without any user interaction. The exploit leverages a buffer overflow in the Windows SMBv1 driver (srv.sys).
MITRE ATT&CK	T1190 (Exploit Public-Facing Application) T1210 (Exploitation of Remote Services)

Evidence 3: meterpreter.dll Delivery

Timestamp	2021-05-08 15:29:58 UTC (Last Write Time: 5/8/2021 3:29 PM)
Frame	590
Filename	meterpreter.dll
File Size	200,262 bytes (195.57 KB)
Source	192.168.1.10 via TCP port 4444
Destination	192.168.1.5
MD5	bb8192f17603c8098f6d9985f1bd1713
SHA256	f471a1873cc2d793e2db1efcc6b0789401c72fdb6b91b842f52569e8b870186
VirusTotal	56 / 72 vendors trojan.meterpreter/shellcode Family: meterpreter, shellcode, marte
Significance	TCP 4444 is the default Metasploit listener port. The file was named meterpreter.dll without obfuscation, confirming the attacker did not attempt to disguise the tooling. 56/72 VirusTotal detections is among the highest possible confidence levels.
MITRE ATT&CK	T1105 (Ingress Tool Transfer) T1571 (Non-Standard Port)

Evidence 4: Process Injection into spoolsv.exe

Timestamps	2021-05-08 15:30:13 UTC and 2021-05-08 15:31:05 UTC
Event ID	8 (Sysmon CreateRemoteThread)
Host	WIN7.seclan.local
Source Process	C:\Windows\System32\spoolsv.exe (PID 1092)
Significance	Meterpreter process migration into the Windows Print Spooler service. spoolsv.exe runs continuously as SYSTEM, providing both privilege elevation (the injected agent inherits SYSTEM) and persistence (the agent survives termination of the original exploit shellcode process).
MITRE ATT&CK	T1055.003 (CreateRemoteThread Injection) T1543 (Create or Modify System Process)

Evidence 5: Post-Exploitation Reconnaissance

samr Protocol	WIN7 (192.168.1.5) to DC (192.168.1.1) for domain user and group enumeration
srvsvc Protocol	WIN7 (192.168.1.5) to DC (192.168.1.1) for share and session enumeration
file.io Connection	TCP 443 from WIN7 to 18.232.47.250 (file.io anonymous file sharing)
Significance	Standard Meterpreter post-exploitation reconnaissance pattern. SAMR enumerates users for credential dumping or lateral movement targeting; SRVSVC enumerates shares for data discovery; file.io connections are typical of either tool staging or initial data exfiltration.
MITRE ATT&CK	T1018 (Remote System Discovery) T1087 (Account Discovery) T1135 (Network Share Discovery) T1567 (Exfiltration Over Web Service)

5. Cross-Host Correlation and Final Findings

5.1 Single Attacker, Two Distinct Vectors

Both compromised hosts were attacked by the same threat actor operating from 192.168.1.10. The attacker chose different initial access vectors tailored to each victim's exposed services and patch level. Against the unpatched Windows 7 SP1 host (192.168.1.5), the attacker exploited the long-known MS17-010 EternalBlue vulnerability for instant kernel-mode remote code execution. Against the better-patched Windows 10 host (192.168.1.3), the attacker chose credential-based access via RDP using the Administrator account, indicating the credentials were either weak or had been previously compromised.

Both attacks ultimately delivered Metasploit Framework payloads. The Suricata signature ET MALWARE Possible Metasploit Payload Common Construct Bind_API fired against both victims, providing strong correlation that the same attacker tooling was used. The two-phase attack against the Windows 10 host (RDP access at 16:15 followed by payload drop at 16:53) suggests the attacker first secured a foothold and then loaded additional tooling once inside, while the Windows 7 attack was a single rapid exploit-and-deliver chain completed in approximately 11 seconds.

5.2 MITRE ATT&CK Mapping Summary

The complete attack chain maps to the following MITRE ATT&CK techniques:

- **T1190 Exploit Public-Facing Application:** via the EternalBlue MS17-010 exploit against 192.168.1.5
- **T1078 Valid Accounts / T1021.001 Remote Desktop Protocol:** RDP login to 192.168.1.3 using compromised Administrator credentials
- **T1105 Ingress Tool Transfer:** delivery of meterpreter.dll and pingthing.exe over TCP 4444 and raw TCP
- **T1036 Masquerading:** the pingthing.exe filename chosen to appear innocuous in the Downloads folder
- **T1055.003 Process Injection:** Meterpreter migration into spoolsv.exe via CreateRemoteThread
- **T1059.003 Windows Command Shell:** Meterpreter Bind_API construct providing interactive shell access
- **T1018 / T1087 / T1135 Discovery:** samr and srvsvc enumeration against the domain controller
- **T1567 Exfiltration Over Web Service:** TCP 443 connections to file.io anonymous sharing service
- **T1571 Non-Standard Port:** TCP 4444 used outside its standard application context

5.3 Indicators of Compromise (IOCs)

Attacker IP	192.168.1.10 (Tu1BOSrnxGuOhpcp, Linux)
File: pingthing.exe	MD5 FC3E1F813E3B23F65AF8682B6E2EA771 / SHA256 1D08CF7E14170A45DE8E98B183142365F80F4D36B32 DF2F638B8473FC1F963D4
File: meterpreter.dll	MD5 bb8192f17603c8098f6d9985f1bd1713 / SHA256

	f471a1873cc2d793e2db1efcc6b0789401c72fdb6b91b842f52569e8b870186
Network: TCP 4444	Metasploit default listener port used to deliver meterpreter.dll to WIN7
Network: file.io	18.232.47.250 (TCP 443) used by compromised WIN7 for staging or exfiltration
File path	C:\Users\Administrator\Downloads\pingthing.exe on WINDOWS10
Process	spoolsv.exe (PID 1092) on WIN7 hosting injected Meterpreter agent
CVE	CVE-2017-0144 (EternalBlue / MS17-010) used against WIN7 SP1 (Build 7601)

5.4 Recommendations

- Apply MS17-010 patch (KB4012212) to all remaining Windows 7 hosts. EternalBlue has been patched since March 2017 and continued exposure represents critical risk.
- Decommission Windows 7 entirely. Microsoft ended extended support in January 2020. Any Windows 7 host on a production network is unsupported and exposes the network to known unpatched vulnerabilities.
- Disable SMBv1 across the environment via Group Policy. SMBv1 is required for the EternalBlue exploit and should be disabled wherever it is not explicitly needed for legacy compatibility.
- Rotate the local Administrator password on all workstations and consider deploying LAPS (Local Administrator Password Solution) to prevent credential reuse across hosts.
- Audit RDP exposure and restrict inbound RDP at the host firewall to only authorized administrative jump hosts. Consider deploying an RDP gateway with multi-factor authentication.
- Block outbound connections to anonymous file-sharing services (file.io, transfer.sh, etc.) at the perimeter firewall.
- Tune the Suricata ruleset to elevate ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request alerts from low to high severity for accounts that should never authenticate via RDP.
- Deploy EDR with behavioral detection. The pingthing.exe payload had zero AV detections but would have triggered behavioral rules for raw-TCP-delivered executables and Metasploit Bind_API patterns.

6. References

- **MITRE ATT&CK Framework (2025):** T1078, T1021.001, T1105, T1036, T1055.003, T1059.003, T1190, T1210, T1571, T1018, T1087, T1135, T1543, T1567. Available at: attack.mitre.org
- **Emerging Threats / Proofpoint (2025):** Suricata rule SID 2025644: ET MALWARE Possible Metasploit Payload Common Construct Bind_API. Emerging Threats Open ruleset signatures for IDS alert generation in Security Onion 2.4.160. Available at: rules.emergingthreats.net
- **VirusTotal (2025):** Hash analysis for pingthing.exe (SHA256: 1D08CF7E14170A45DE8E98B183142365F80F4D36B32DF2F638B8473FC1F963D4) and

meterpreter.dll (SHA256: f471a1873cc2d793e2db1efcc6b0789401c72fdb6b91b842f52569e8b870186). Available at: [virustotal.com](https://www.virustotal.com)

- **Microsoft Security Bulletin MS17-010 (2017):** Critical vulnerability in Windows SMBv1 server (srv.sys). CVE-2017-0144 through CVE-2017-0148. Patched via KB4012212. Available at: docs.microsoft.com/security-updates/securitybulletins/2017/ms17-010
- **Netresec / NetworkMiner (2025):** NetworkMiner 2.8.1 Network Forensic Analysis Tool documentation. Used for host enumeration, file extraction, session mapping, and DNS analysis. Available at: netresec.com
- **Security Onion Solutions (2025):** Security Onion 2.4.160 documentation: Hunt query interface, Alerts dashboard, EVTX import, and Suricata configuration. Available at: securityonionsolutions.com
- **Sysinternals / Microsoft (2025):** Windows Sysmon documentation: Event ID 8 (CreateRemoteThread), Event ID 12 to 14 (Registry), Event ID 15 (FileCreateStreamHash). Available at: docs.microsoft.com/sysinternals/downloads/sysmon
- **Wireshark Foundation (2025):** Wireshark display filter reference, TLS/SSL analysis, and TCP stream reconstruction documentation. Available at: [wireshark.org](https://www.wireshark.org)
- **Rapid7 (2025):** Metasploit Framework default listener configuration (TCP 4444), Bind_API payload construct, and process migration techniques (post/multi/manage/migrate). Available at: docs.metasploit.com